

Epilepsy syndromes (electroclinical syndromes) are well-recognized groupings of clinical (seizure types) and EEG features that occur together. Each syndrome typically shares a common age of onset, deficits (intellectual dysfunction), treatment, and prognosis. Syndromes are classified based on their onset, epilepsy type (focal, generalized or mixed) and development of epileptic encephalopathy (disorder in which epileptic activity contributes to severe impairments in cognition and behavior) *.

Relatively benign syndromes are typically associated with focal, generalized tonic clonic (GTC), typical absences, and myoclonic seizures. Epileptic encephalopathies are typically associated with atonic, tonic, atypical absences and epileptic spasms in addition to the above seizure types [1].

Epilepsy Syndromes

Neonatal Period

- benign neonatal convulsions
- early infantile epileptic encephalopathy (Ohtahara's syndrome) *
- early myoclonic encephalopathy (EME) *

Infancy

- benign infantile seizures
- epilepsy of infancy with migrating focal seizures (migrating partial seizures of infancy)
- myoclonic epilepsy in infancy
- infantile spasms (West syndrome) *
- myoclonic encephalopathy in nonprogressive disorders *

Childhood

- benign epilepsy with centrotemporal spikes (BECTS)
- early childhood onset occipital epilepsy (Panayiotopoulos syndrome)
- late childhood onset occipital epilepsy (Gastaut syndrome)
- childhood absence epilepsy (CAE)
- epilepsy with myoclonic absences (EMA)
- myoclonic-atonic/astatic epilepsy (Doose syndrome)

- generalized epilepsy with febrile seizures plus (GEFs plus)
- severe myoclonic epilepsy (Dravet syndrome) *
- epileptic encephalopathy with continuous spike and wave during sleep (CSWS) *
- Landau Kleffner syndrome (LKS) *
- Lennox–Gastaut syndrome (LGS) *

Adolescence and Adulthood

- autosomal-dominant nocturnal frontal lobe epilepsy (ADNFLE)
- autosomal-dominant temporal lobe epilepsy with auditory features (ADTLE)
- familial temporal lobe epilepsies
- juvenile absence epilepsy (JAE)
- juvenile myoclonic epilepsy (JME)
- epilepsy with generalized tonic clonic seizures alone (EGTCA)
- progressive myoclonic epilepsies (PME)

Any Age (Less Age Specific)

- familial focal epilepsy with variable foci (FFEVF)
- reflex epilepsies

Some epilepsy disorders may not be syndromic but nevertheless share recognizable features (distinctive constellations), these include Rasmussen's syndrome, gelastic seizures with hypothalamic hamartoma (HH), and hemi convulsion-hemiplegia epilepsy (HHE).

The remainder of this chapter discusses the EEG features of common epilepsy syndromes.

Neonatal Syndromes

Benign Neonatal Convulsions

These are also called benign idiopathic neonatal seizures or fifth day fits. They occur in the first week of life as unifocal clonic seizures with apnea/cyanosis. There are no neurodevelopmental deficits. The etiology is believed to be genetic such as potassium channel defects. Typically resolve by the second week (self-limited) though some develop seizures in later life. Interictal EEG may be normal or show “theta pointu alternant” which are bursts of focal, nonreactive theta activity intermixed with multifocal sharp waves that alternate between hemisphere [2].

Early Infantile Epileptic Encephalopathy (Otoharu Syndrome)

This is an epileptic encephalopathy that typically occurs within the first three months of life. It is characterized by tonic seizures, infantile spasms, and

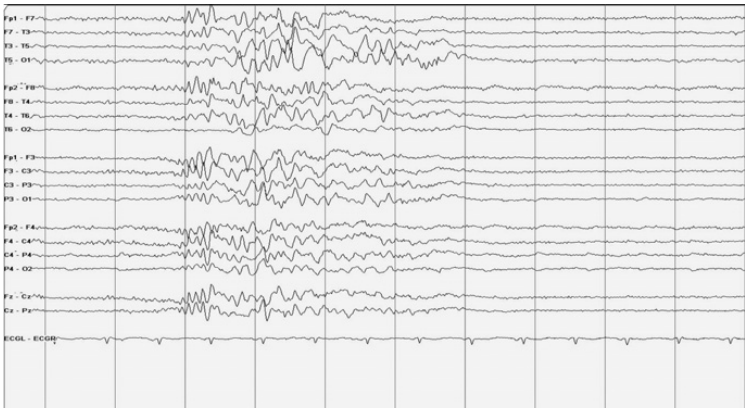


Figure 21.1 One-month-old boy with porencephaly, burst-suppression pattern.

myoclonic seizures. Structural etiologies (such as cortical dysplasia) are common. The prognosis is usually poor with high mortality. Interictal EEG shows burst-suppression regardless of clinical state as shown in Figure 21.1 [3].

Early Myoclonic Encephalopathy (EME)

This is an epileptic encephalopathy typically occurs within the first month of life. It is characterized by myoclonic seizures involving face or limbs, focal seizures, and tonic spasms. Metabolic etiologies (such as nonketotic hyperglycinemia/pyridoxine deficiency) are common. The prognosis is poor with high mortality. Interictal EEG shows burst-suppression, may be only during sleep. At 3–5 months of age, the waking EEG is replaced by hypsarrhythmia, this reverts to burst-suppression around 2–3 years of age [3].

Infantile Syndromes

Benign Infantile Seizures

Both familial and nonfamilial forms exist, nonfamilial forms. They typically occur between 3 months to 3 years of life (nonfamilial forms have later onsets) with focal clonic seizures of the head, face, and limbs that may generalize. Normal Neurodevelopment. Self-limited with a good prognosis. Interictal EEG is normal. Ictal EEG shows focal onset temporal/posterior seizures [4].

Epilepsy of Infancy with Migrating Focal Seizures

These were previously called migrating partial seizures of infancy. Typical onset is around 3 months of age with sporadic focal motor seizures. These often cluster and show bilateral spread. Neurodevelopment is normal at presentation but declines. Genetic etiologies (such as sodium and potassium channel defects) are common. Poor prognosis with high mortality. Interictal EEG shows multifocal slowing with disruption of sleep architecture. Ictal EEG shows multifocal or migrating seizure onsets [5].

Myoclonic Epilepsy in Infancy

This is the earliest form of idiopathic generalized epilepsy occurring between 3 months to 3 years of age. Bilateral myoclonic jerks, head drops, or truncal spasms may also occur. Seizures may be reflexive to auditory or other stimuli. Neurodevelopment is normal. Self-limited with good prognosis. Jerks are associated with fast generalized, irregular spike, or polyspikes discharges as shown in Figure 21.2 [6].

Infantile Spasms (West Syndrome)

West syndrome is a triad of spasms, hypsarrhythmia on EEG, and developmental delay. The typical onset is around 4 to 8 months of age with clusters of flexor or extensor spasms several times a day. Structural (such as tuberous sclerosis), genetic, metabolic, and infectious etiologies are common. Epileptic encephalopathy with a poor prognosis. Hypsarrhythmia refers to continuous,

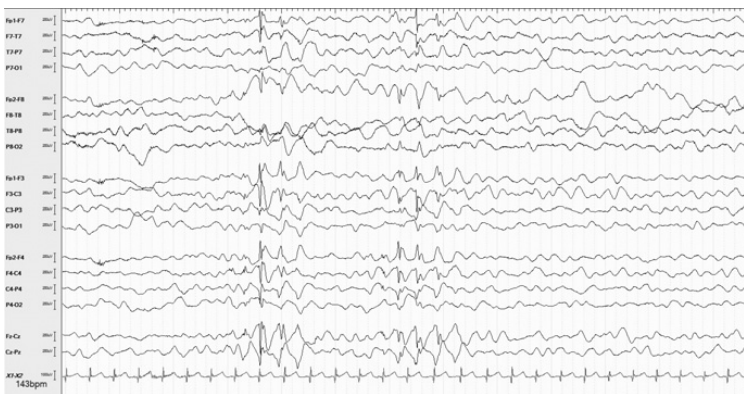


Figure 21.2 Twelve-month-old boy with myoclonic epilepsy in infancy, generalized 3 Hz discharges.

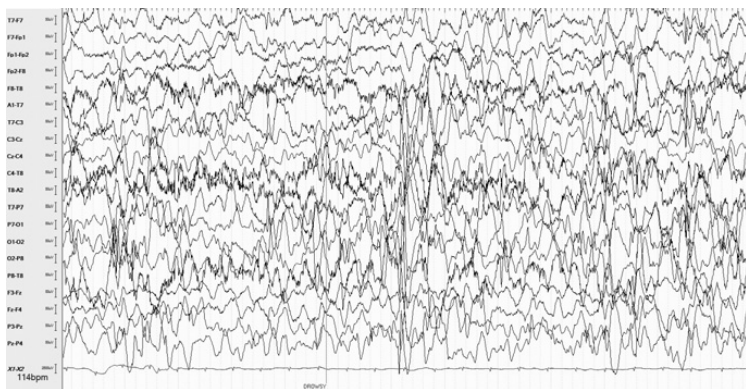


Figure 21.3 Nine-month-old boy with West syndrome, hypsarrhythmia (sensitivity 10 μ V).

arrhythmic, high-amplitude, asynchronous delta activity with multifocal independent discharges (chaotic) on interictal EEG as shown in Figure 21.3. Diffuse suppressions are seen during spasms as shown in Chapter 19, Figure 19.6 [3,7].

Myoclonic Encephalopathy in Nonprogressive Disorders

This is a rare disorder with typical onset in the first year of life and is characterized by continuous myoclonic jerking in the absence of a progressive encephalopathy. Genetic etiology such as Angelman syndrome are common. Epileptic encephalopathy with poor prognosis. Interictal EEG shows diffuse theta-delta or delta with spikes as shown in Figure 21.4 [3].

Childhood Syndromes

Benign Epilepsy with Centrotemporal Spikes (BECTS)

This is also called Rolandic epilepsy and is the most common focal epilepsy of childhood. Typically occur around 7 to 10 years of age with unilateral or bilateral, sensory or motor seizures that are characterized by oropharyngeal manifestations (numbness, paresthesia, gurgling, grunting, or drooling) or speech arrest. May spread to face and hand with secondary generalization. Seizures are commonly nocturnal. Neurodevelopmentally normal, though occasional concerns with language may occur. The etiology is believed to be genetic. Good prognosis with most patients remitting before age 16. Interictal EEG typically shows centrotemporal spikes (CTS) these are clusters of high-amplitude, sharp and slow-wave

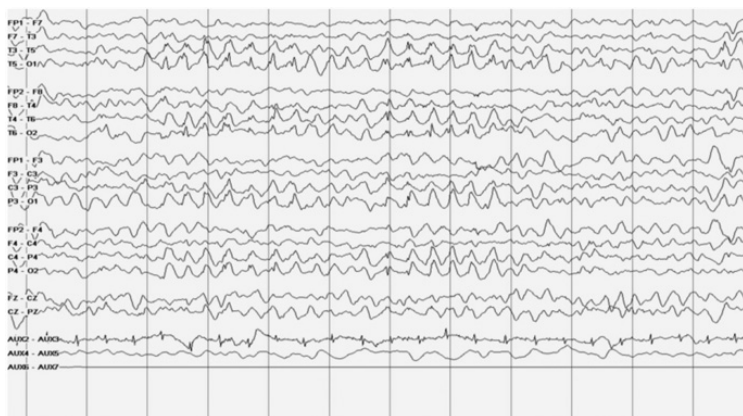


Figure 21.4 Twelve-month-old girl with Angelman's syndrome, occipital-predominant delta with spikes.

discharges that that phase reverse over central (C3-C4), temporal, or parietal regions on a longitudinal bipolar montage. They can be amplified during sleep and sometimes with somatosensory stimulation. They are usually bilateral and asymmetric with characteristic horizontal dipoles (surface negative over the central region but simultaneously positive over the frontal region). Slowing after spikes may suggest a greater challenge with seizure control. Decrease of spikes in sleep, fast/polyspikes, and burst-suppression after spikes may suggest alternative diagnosis. Ictal EEG is variable and may show evolving discharges, low-voltage fast rhythmic activity, or rhythmic theta. Figure 21.5 shows centro-temporal spikes and Figure 21.6 shows their characteristic horizontal dipole on referential montage [8].

Early Childhood Onset Occipital Epilepsy (Panayiotopoulos Syndrome)

Typically occurs between 3 to 6 years of age with agitation, headache and autonomic symptoms (such as a nausea, vomiting, pallor, and cyanosis) leading to a prolonged hemiclonic or generalized motor seizure. Autonomic status epilepticus may occur. Seizures are infrequent and may not need treatment. Neurodevelopmentally normal. Self-limited epilepsy with good prognosis. Interictal EEG shows salvos of bilateral, independent, occipital spikes, especially in sleep. Figure 21.7 shows occipital spikes in Panayiotopoulos syndrome [9].

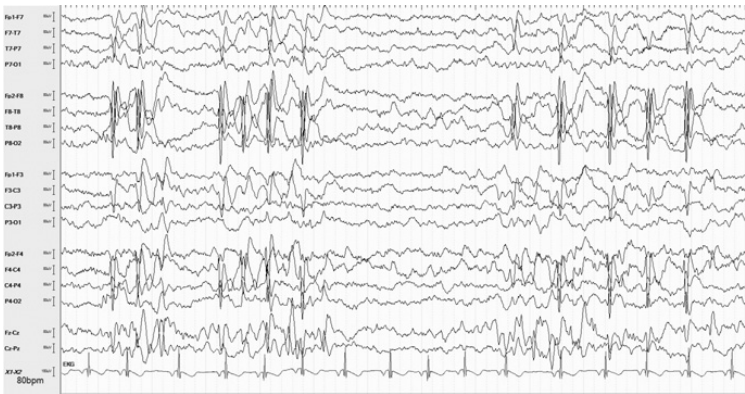


Figure 21.5 Seven-year-old girl with Rolandic epilepsy, centrotemporal spikes with a right temporal predominance (sensitivity 20 μ V).

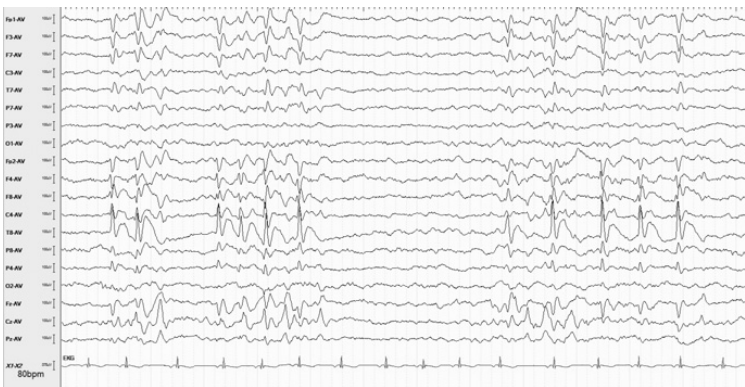


Figure 21.6 Same patient as in Figure 21.5, horizontal dipole on referential montage (sensitivity 20 μ V).

Late Childhood Onset Occipital Epilepsy (Gastaut Syndrome)

Typically occurs between 8 and 11 years of age with elementary visual auras or visual field cuts leading to a focal seizure with rare secondary generalization. Headaches occur but autonomic symptoms are uncommon. Seizures are more frequent compared to Panayiotopoulos syndrome, needing treatment.

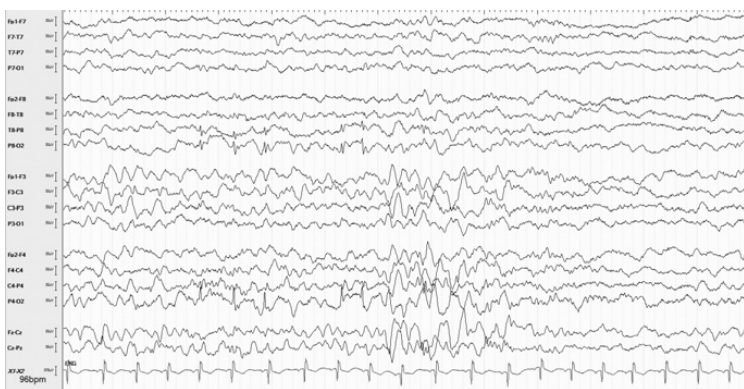


Figure 21.7 Three-year-old girl with Panayiotopoulos syndrome, right occipital spikes (sensitivity 10 μ V).

Neurodevelopmentally normal. Self-limited epilepsy with good prognosis. Interictal EEG shows bilateral, independent, occipital spikes in salvos, especially with sleep and eye closure [9].

Childhood Absence Epilepsy (CAE)

Begins in the preschool (3 to 10 years) with typical absences (staring spells). These are characterized by transient unresponsiveness and event amnesia that may occur several times a day. Seizures can be provoked by hyperventilation and often have motor automatisms. Minority have at least one generalized tonic clonic seizure. Genetic etiologies are thought to be responsible for this syndrome. In those with an early onset Glucose Transporter (GLUT-1) deficiency should be considered. Neurodevelopmentally normal. Self-limited epilepsy but often poor social outcomes. Interictal EEG is typically normal. Ictal EEG shows regular 3 Hz generalized spike and wave pattern as shown in Figure 21.8. Rarely photosensitive [10].

Epilepsy with Myoclonic Absences (EMA)

Typical onset occurs around 5 to 12 years of age with myoclonic absences. These are brief absences (10 to 60 s) with axial hypertonía. Drop attacks or generalized tonic clonic seizures may also occur. Neurodevelopmental delay/decline is common. Both structural or genetic etiologies may be responsible. The prognosis is variable but cognitive impairment and epilepsy often persist into adulthood. Interictal EEG may be normal or show generalized discharges.

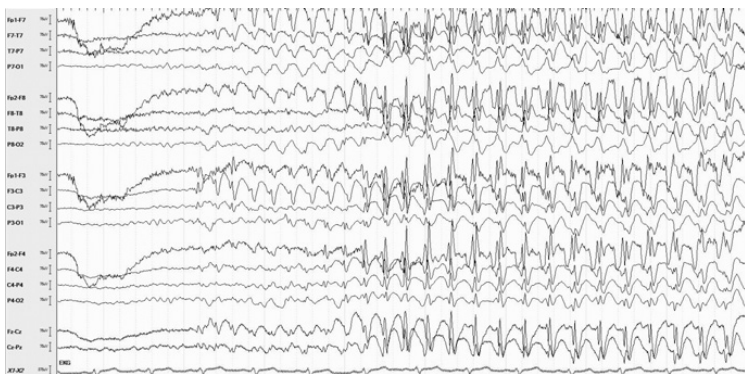


Figure 21.8 Eight-year-old boy with CAE, typical absences with 3 Hz generalized spike and wave discharges.

Ictal EEG shows regular 3 Hz generalized spike and wave discharges typically provoked by hyperventilation or photic stimulation [11].

Myoclonic-Atonic/astatic Epilepsy (Dooze Syndrome)

Typically occurs around 1 to 5 years of age with acute onset of atonic, absence, and myoclonic seizures in a previously normal child. There may be a family history of epilepsy. Cognitive decline occurs and seizures are refractory to medications. Interictal EEG may be normal or show a central rhythmic theta. Ictal EEG (myoclonic-atic seizure) shows 2–4 Hz generalized spike wave discharges as shown in Figure 21.9 [12].

Generalized Epilepsy with Febrile Seizures Plus (GEFS Plus)

Febrile seizures persist beyond 6 years of age or occur before six months. There is a strong family history of generalized epilepsy and/or febrile seizures. Nonfebrile seizures may occur. The etiology is thought to be secondary to a sodium channel related gene defect (SCN1A). Normal neurodevelopment. Self-limited, usually remits before puberty. Interictal EEG is normal. Ictal EEG depends on the seizure type [13].

Severe Myoclonic Epilepsy (Dravet Syndrome)

Epileptic encephalopathy with a typical onset before 2 years of age, heralded by a prolonged febrile seizure (hemi convulsion). These are followed by focal afebrile seizures (switch sides), myoclonus, and later multiple seizure types

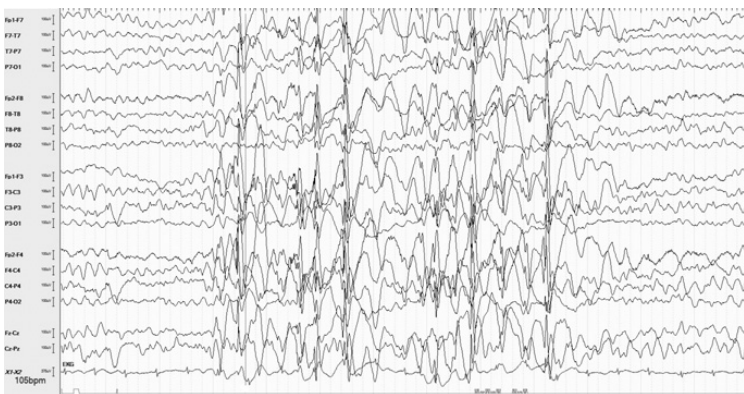


Figure 21.9 Three-year-old boy with Dose syndrome (myoclonic astatic epilepsy), 3 to 2.5 Hz generalized spike and wave discharges (sensitivity 20 μ V).

often precipitated by fever. Genetic etiology (SCN1A defect) is responsible. Neurodevelopment may be normal at presentation, but progressively declines. Seizures are refractory and the prognosis is poor. Interictal EEG is initially normal, then deteriorates, and generalized discharges appear as shown in Figure 21.10 [3].

Epileptic Encephalopathy with Continuous Spike and Wave during Sleep (CSWS)

This is also called electrical status epilepticus during slow-wave sleep (ESES). Onset occurs between ages 5 to 10 years of age with seizures followed by cognitive decline that persists despite seizure resolution. Both structural and genetic etiologies may be responsible. EEG shows generalized discharges, marked sleep activation comprising greater than 85% of non-rapid eye movement (NREM) sleep as shown in Figure 21.11. Discharges significantly reduce in REM sleep. Typically, poor prognosis with severe behavioral and intellectual dysfunction [3].

Landau Kleffner Syndrome (LKS)

Typically, onset occurs between 3 to 6 years of age with verbal auditory aphasia, expressive aphasia, and focal seizures follow. Genetic (GRIN2A mutation) or autoimmune etiologies may be responsible. Seizures remit but language/cognitive deficits worsen (epileptic encephalopathy). EEG is like ESES but may have focal/multifocal spikes [3].



Figure 21.10 Twelve-month-old boy with Dravet syndrome (severe myoclonic epilepsy), background slowing, and abundant generalized discharges.



Figure 21.11 Seven-year-old girl with intellectual disability, continuous generalized discharges during NREM sleep.

Lennox–Gastaut Syndrome (LGS)

This is an epileptic encephalopathy with onset between 1 to 10 years of age. Multiple seizure types including tonic, atypical absences, and atonic

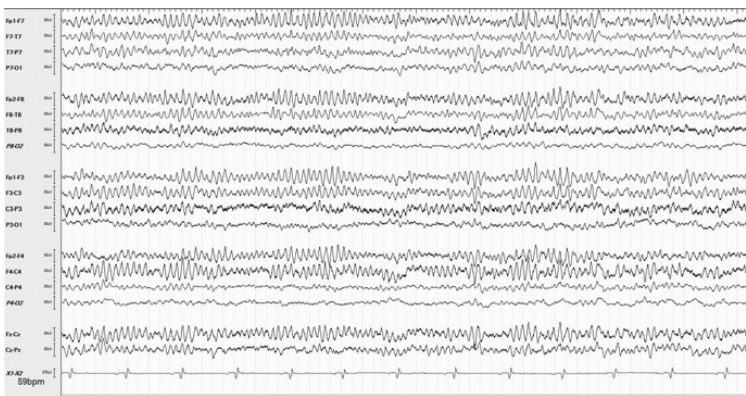


Figure 21.12 Thirty-one-year-old woman with LGS, generalized bursts of 10 Hz activity during NREM sleep.

seizures are characteristic. Myoclonic, focal, and generalized tonic clonic seizures also occur. In some, infantile spasms precede the onset of LGS. Structural, postinfectious, or genetic etiologies are usually responsible. The prognosis is poor with cognitive dysfunction, refractory epilepsy, and status epilepticus. Interictal EEG shows slow background with bursts of irregular generalized slow spike and wave discharges that are accentuated during NREM sleep as shown in Chapter 14, Figure 14.11. Generalized fast bursts (10 Hz) during NREM sleep also occur (Figure 21.12). Ictal EEG depends on seizure type, may show generalized discharge followed by a burst of diffuse fast activity (tonic seizures), generalized slow spike and wave discharges (atypical absences), polyspikes or diffuse fast activity (atonic or myoclonic seizures) as previously described in Chapter 19 [3,12].

Adolescent and Adult Syndromes

Autosomal-Dominant Nocturnal Frontal Lobe Epilepsy (ADNFLE)

Typically occurs around 12 years of age with nocturnal hyperkinetic seizures that cluster. Retained awareness and aura are common. Normal neurodevelopment. Genetic etiologies (nicotinic cholinergic receptor defects) are responsible. Interictal EEG is normal. Ictal EEG shows bifrontal onsets [14].

Autosomal-Dominant Temporal Lobe Epilepsy with Auditory Features (ADTLE)

This is also called autosomal-dominant lateral temporal lobe epilepsy. The onset usually occurs in the first or second decade of life, but may be later. Lateral temporal lobe seizures with prominent ictal auditory symptoms and/or receptive aphasia occur. Normal neurodevelopment. Genetic etiologies (LGII defect) are responsible. Interictal EEG is normal. Ictal EEG shows lateral temporal lobe onset. Pharmacoresponsive epilepsy with good prognosis [15].

Familial Mesial Temporal Lobe Epilepsies (FMTLE)

Numerous relatively benign, familial, mesial temporal lobe epilepsies occur typically in the first or second decade of life and follow a mild clinical course that may remit. Three main presentations include those without febrile seizures/hippocampal sclerosis (prominent autonomic and psychic aura), those with hippocampal sclerosis (prominent automatisms and postictal changes) and those with febrile seizures. Hippocampal sclerosis may occur in asymptomatic family members [16].

Juvenile Absence Epilepsy (JAE)

Typical onset is around 10 to 12 years of age with infrequent absences followed by generalized tonic clonic seizures. Cognitive dysfunction may occur. Variable prognosis, typically pharmacoresponsive but may not remit with age. Interictal EEG shows 3–6 Hz generalized discharges not typically provoked by hyperventilation (Figure 21.13) [10].

Juvenile Myoclonic Epilepsy (JME)

Most common genetic generalized epilepsies with typical onset between ages 12 to 20 years. Classically presents with morning myoclonus, generalized tonic clonic seizures, and less commonly absences. Seizure triggers include sleep deprivation, alcohol, and photic stimulation. Good prognosis with normal cognition and pharmacoresponsive seizures but typically doesn't resolve with age. EEG shows 4–6 Hz (fast) generalized discharges (Figure 21.14) [10].

Epilepsy with Generalized Tonic Clonic Seizures Alone

Typically occur between 10 to 25 years with generalized tonic clonic seizures upon awakening. Some may have nocturnal seizures. Sleep deprivation, alcohol, and photosensitivity are common triggers. Cognition is normal. Interictal EEG shows generalized discharges [10].



Figure 21.13 Ten-year-old boy with JAE, generalized 4 Hz spike, and wave discharges.

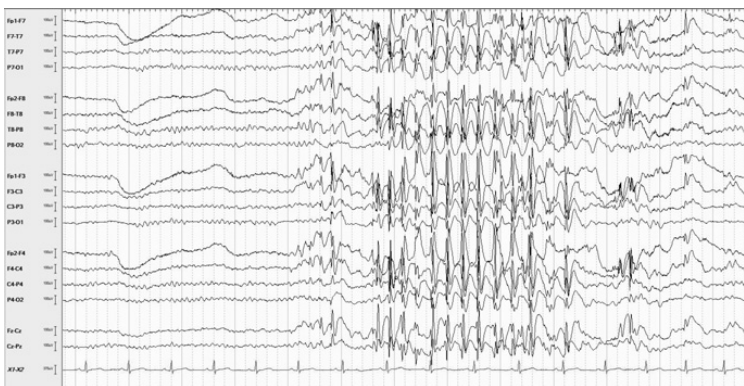


Figure 21.14 Twelve-year-old boy with JME, generalized 4 Hz spike, and wave discharges (sensitivity 20 μ V).

Progressive Myoclonic Epilepsies

These are a manifestation of underlying neurometabolic and neurodegenerative diseases. They are characterized by cognitive decline (epileptic encephalopathy)

and intractable myoclonus. Unvericht–Lundborg syndrome (UVL), Lafora body disease, and neuronal ceroid lipofuscinosis (NCL) are common causes. EEG typically shows a deteriorated background, multifocal epileptic discharges, and photosensitivity with giant evoked potentials. Poor prognosis with high mortality [3].

Less Age-Specific Syndromes and Other Constellations

Familial Focal Epilepsy with Variable Foci (FFEVF)

Typically occurs in the first or second decade of life and is characterized by focal onset seizures from different cortical regions. Variable neurodevelopmental outcomes including autism may occur. Genetic etiology (DEPDC5 gene defect on chromosome 22). EEG findings are variable [17].

Progressive Unilateral Encephalopathy of Childhood (Rasmussen's Syndrome)

Typically occurs between 3 and 14 years of age with focal seizures (often *epilepsia partialis continua*) and other unilateral deficits. There is progressive unihemispheric cortical atrophy on neuroimaging. Normal development at onset but later declines. Histopathology shows a T-cell mediated encephalitis (autoimmune etiology) with microglial nodules and reactive astrogliosis. Interictal EEG shows unihemispheric slowing and epileptic discharges [18].

Gelastic Seizures with Hypothalamic Hamartoma (HH)

Hypothalamic hamartomas are intrinsically epileptogenic due to GABA neurons. They classically present with gelastic seizures (emotionless laughter). Other seizures including dacrystic seizures (crying) may occur. Precocious puberty, behavioral problems, and cognitive dysfunction are common. Interictal and ictal EEG may be normal [19].

Hemi Convulsion-Hemiplegia Epilepsy (HHE)

This is an uncommon form of epilepsy that occurs in late infancy or toddlers and is characterized by prolonged focal or generalized status epilepticus followed by hemiplegia. May be idiopathic or symptomatic due to other pathology (such as stroke, infection, or trauma) [20].

Reflex Epilepsies

Multiple reflex epilepsies have been identified based on the triggering stimulus such as visual/photosensitivity, reading, music, tactile, hot water, and others.

Chapter Summary

Common age-dependent epilepsy syndromes

Syndrome	Epilepsy	Seizures	EEG (interictal)	Cognitive impairment	Course
Neonatal syndromes					
Benign neonatal convulsions	Focal	Focal (apneic)	Theta pointu alternant	None	Self-limited
Otohara syndrome	Epileptic encephalopathy	Tonic	Burst-suppression	Severe	Poor prognosis
Early myoclonic encephalopathy	Epileptic encephalopathy	Myoclonic	Burst-suppression	Severe	Poor prognosis
Infantile syndromes					
Benign infantile seizures	Focal	Focal clonic	Normal	None	Self-limited
Epilepsy of infancy with migrating focal seizures	Multifocal	Clusters of focal motor	Multifocal or migrating patterns	Initially normal, then declines	Poor prognosis
Myoclonic epilepsy in infancy	Generalized (IGE)	Myoclonus, head drops, truncal spasms	1–3 Hz generalized discharges	None	Self-limited
West syndrome	Epileptic encephalopathy	Clusters of flexor or extensor spasm	Hypsarrhythmia	Severe	Poor prognosis
Myoclonic encephalopathy in nonprogressive disorders	Epileptic encephalopathy	Myoclonus	Diffuse theta or delta with spikes	Severe	Poor prognosis
Childhood syndromes					
Benign epilepsy with centrotemporal spikes	Focal (Rolandic)	Oropharyngeal, speech or autonomic	Centrotemporal spikes	None	Self-limited

(cont.)

Syndrome	Epilepsy	Seizures	EEG (interictal)	Cognitive impairment	Course
Panayiotopoulos syndrome	Focal occipital	Autonomic, focal motor	Occipital spikes	None	Self-limited
Gastaut syndrome	Focal occipital	Visual, focal motor	Occipital spikes	None	Self-limited
Childhood absence epilepsy	Generalized (IGE)	Typical absences	GSW (3 Hz)	None	Self-limited
Epilepsy with myoclonic absences	Generalized (IGE)	Absences with axial hypertonia	GSW (3 Hz)	Moderate	Variable
Doose syndrome	Generalized (IGE)	Atonic, absences and myoclonic	Central rhythmic theta, GSW (2–4 Hz)	Moderate	Poor prognosis
GEF plus	Generalized	Febrile and afebrile seizures	Normal	None	Self-limited
Dravet syndrome	Epileptic encephalopathy	Febrile and afebrile seizures, other types	Slowing, generalized discharges	Initially normal, then severe decline	Poor prognosis
CSWS	Epileptic encephalopathy	Generalized	Generalized discharges, 85% of NREM sleep	Severe	Poor prognosis
LKS	Epileptic encephalopathy	Focal	as above	Language	Poor prognosis
LGS	Epileptic encephalopathy	Tonic, atypical absences, atonic and others	Slow spike and wave (2–2.5 Hz)	Severe	Poor prognosis
Adolescent and adult syndromes					
ADNFLE	Focal	Hyperkinetic	Normal	None	Pharmacoresponsive

(cont.)

Syndrome	Epilepsy	Seizures	EEG (interictal)	Cognitive impairment	Course
ADTLE	Focal	Auditory aura, receptive aphasia	Normal	None	Pharmaco-responsive
FMTLE	Focal	Autonomic, psychic aura	Normal	None	Pharmaco-responsive
JAE	Generalized (IGE)	Infrequent absences, GTCs	Generalized discharges (3.5–6 Hz)	Variable	Variable
JME	Generalized (IGE)	Myoclonus, GTCs, rare absences	Generalized discharges (4–6 Hz)	None	Pharmaco-responsive
Epilepsy with generalized tonic clonic seizures alone	Generalized (IGE)	GTC	Generalized discharges	None	Pharmaco-responsive
PME	Epileptic encephalopathy	Myoclonus	Giant photic evoked potentials	Severe	Poor prognosis

References

1. Scheffer IE, Berkovic S, Capovilla G, et al. ILAE classification of the epilepsies: position paper of the ILAE Commission for Classification and Terminology. *Epilepsia*. 2017 Apr;**58**(4):512–21.
2. Miles DK, Holmes GL. Benign neonatal seizures. *Journal of Clinical Neurophysiology*. 1990 Jul;**7**(3):369–79.
3. Donat JF. Topical review article: the age-dependent epileptic encephalopathies. *Journal of Child Neurology*. 1992 Jan;**7**(1):7–21.
4. Vigeveno F. Benign familial infantile seizures. *Brain and Development*. 2005 Apr **1**;**27**(3):172–7.
5. Caraballo RH, Fontana E, Darra F, et al. Migrating focal seizures in infancy: analysis of the electroclinical patterns in 17 patients. *Journal of Child Neurology*. 2008 May;**23**(5):497–506.
6. Dravet C, Bureau M. Benign myoclonic epilepsy in infancy. *Epileptic Syndromes in Infancy, Childhood and Adolescence*. 2005;**4**:77–88.

7. Quigg M. *EEG pearls*. Mosby Elsevier, New York; 2006.
8. Donald Shields W, Carter Snead O III. Benign epilepsy with centrotemporal spikes. *Epilepsia*. 2009 Sep;**50**:10–15.
9. Yalçın AD, Kaymaz A, Forta H. Childhood occipital epilepsy: seizure manifestations and electroencephalographic features. *Brain and Development*. 1997 Sep 1;**19**(6):408–13.
10. Mattson RH. Overview: idiopathic generalized epilepsies. *Epilepsia*. 2003 Mar;**44**:2–6.
11. Tassinari CA, Rubboli G, Gardella EL, Michelucci RO. Epilepsy with myoclonic absences. *Epilepsy in Children*. 2004 Feb;**27**:189–94.
12. Stephani U. The natural history of myoclonic astatic epilepsy (Doose syndrome) and Lennox-Gastaut syndrome. *Epilepsia*. 2006 Nov;**47**:53–5.
13. Camfield P, Camfield C. Febrile seizures and genetic epilepsy with febrile seizures plus (GEFS+). *Epileptic Disorders*. 2015 Jun;**17**(2):124–33.
14. Bertrand D, Picard F, Le Hellard S, et al. How mutations in the nAChRs can cause ADNFLE epilepsy. *Epilepsia*. 2002 Jun;**43**:112–22.
15. Michelucci R, Pasini E, Nobile C. Lateral temporal lobe epilepsies: clinical and genetic features. *Epilepsia*. 2009 May;**50**:52–4.
16. Gambardella A, Labate A, Giallonardo A, Aguglia U. Familial mesial temporal lobe epilepsies: clinical and genetic features. *Epilepsia*. 2009 May;**50**:55–7.
17. Dibbens LM, De Vries B, Donatello S, et al. Mutations in DEPDC5 cause familial focal epilepsy with variable foci. *Nature Genetics*. 2013 May;**45**(5):546.
18. Vining EP, Freeman JM, Brandt J, Carson BS, Uematsu S. Progressive unilateral encephalopathy of childhood (Rasmussen's syndrome): a reappraisal. *Epilepsia*. 1993 Jul;**34**(4):639–50.
19. Arroyo S, Lesser RP, Gordon B, et al. Mirth, laughter and gelastic seizures. *Brain*. 1993 Aug 1;**116**(4):757–80.
20. Gastaut H, Poirier F, Payan H, Salamon G, Toga M, Vigouroux MH. HHE syndrome hemiconvulsions, hemiplegia, epilepsy. *Epilepsia*. 1959 Jan;**1**(1–5):418–47.